Assignment CO1-AT2

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Machine Learning - ITA0614

1Q) Design a machine Learning model for spam email classification using labeled data.

(a) Explain the choice of algorithm (eg. Naive Bayes, SVM, or Logistic Regression). (b) Describe the preprocessing steps such as tokenization, stop word removal, and vectorization. (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

A) (a) choice of Algorithm

* Naive Bayes is commonly used for spam email classification because:
  - It works well with text data.
  - It is fast and computationally efficient
  - It assumes independence between words, making classification simple.
  - It performs well even with large datasets.

* SVM or Logistic Regression can also be used, but Naive Bayes is
* preferred for text classification.

(b) Preprocessing steps

1. Tokenization - split email text into individual words (tokens)

2. Lowercasing - Convert all text to lowercase.

3. Stopword Removal - Remove common words like, the, is, and.

4. Stemming / Lemmatization - Remove common words like the, is, and. Convert words to their root from (e.g. running → run).
5. Vectorization - Convert text into numerical from using Bag of words (BoW) or TF-IDF.

(c) Evaluation Metrics :

Accuracy : Precentage of correctly classified emails.

Precision : Percentage of predicted spam emails that are actually spam

Recall : Percentage of actual spam emails correctly identified.

F1-score : Harmonic mean of precision and reccall.

Conclusion:

* In Design of Machine Leaning model for spam email classification can accurately distinguish spam and non-spam emails by training on labeled data.
* Using Naive Bayes along with preprocessing techniques such as tokenization, stop-word removal, and TF-IDF/vectorization improves classification performance.

2) In the context of the candidata - Elimination in Algorithm, consider a dataset aith attributes: sky, AirTemp, Humidity, wind, water, forecast and target concept Enjoysport.

a) Explain the initialization of S (specific boundary) and G (general boundary).

b) show how S and G change with each training Example.

c) compare Candidate - Elimination with the find-S algorithm in terms of generalization capability.

(a) Initialization of S and G

- Specific Boundary (S):
  starts with most specific hypothesis
  Example : $S < \{<\phi,\phi,\phi,\phi,\phi,\phi>\}$
- General Boundary (G):
  starts with most general hypothesis
  Example : $G = \{<?,?,?,?,?,?>\}$

(b) Changes in S and G with Training Examples.

Suppose the training examples are :

Example	Sky	AirTemp	Humidity	wind	water	forecast	Enjoysport
1.	Sunny	warm	Normal	strong	warm	same	yes
2.	Sunny	warm	High	Strong	warm	same	yes
3.	Rainy	cold	High	strong	warm	change	No

After Example 1 (positive):

S = <Sunny, warm, Normal, Strong, warm, same>

G = <?, ?, ?, ?, ?, ?>

After Example 2 (positive):

S = <Sunny, warm, ?, strong, warm, same>

G = <?, ?, ?, ?, ?, ?>

After Example 3 (Negative):

S = <Sunny, warm, ?, strong, warm, same>

G = { <Sunny, ?, ?, ?, ?, ?>,
< ?, warm, ?, ?, ?, ?>,
< ?, ?, ?, ? ?, same>
}

* The Algorithm continues updating S and G until the version space is obtained.

c) Candidate - Elimination vs Find-S

Candidate Elimination	Find-S
Maintains both S and G boundaries	Maintains only the most specific hypothesis
Learns from positive and Negative examples	uses only positive examples
produces all consistent hypothesis (version space)	Produces a single hypothesis
Better generalization capability	Limited Generalization
More computationally expensive	Simple and fast

* Candidate Elimination is more powerful because it maintains all hypothesis consistent with the training data, S - learns only one most specific hypothesis.